# ForteBank Vision

What is achievable in 48 hours with 42 labeled examples, no GPU, no cloud, and a legal constraint that the model must be explainable.

**Nazarbayev University Datathon 2026**

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# The constraints define the problem

*This is not a paper. It is a feasibility study under four simultaneous constraints.*

### DATA

- 42 labeled clips (22 positive, 20 negative)
- 10 to 73 seconds per clip
- No bounding boxes, one timestamp per violation

### COMPUTE

- 48 hours, one M1 laptop
- No GPU at inference (branch server target)
- Deployment target: $5/month VPS

### LEGAL

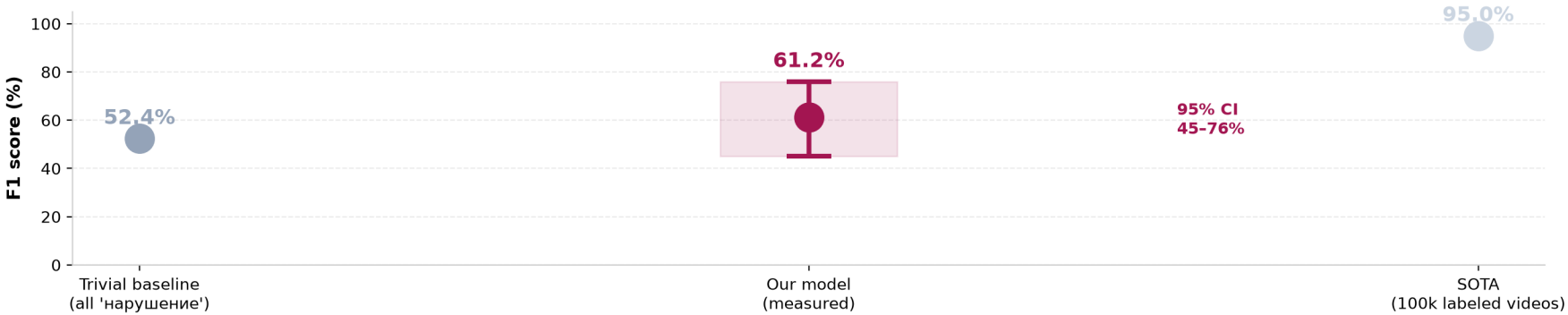
- No cloud inference (data residency)
- Every output cites an RK normative clause
- No hallucinated citations

### EVALUATION

- Test set filmed from a different camera angle
- No published baseline for this task
- F1 is the target metric

# Statistical validity: n=42 spans 30 F1 points

*Every number on subsequent slides inherits this uncertainty.*




## WHAT THIS MEANS

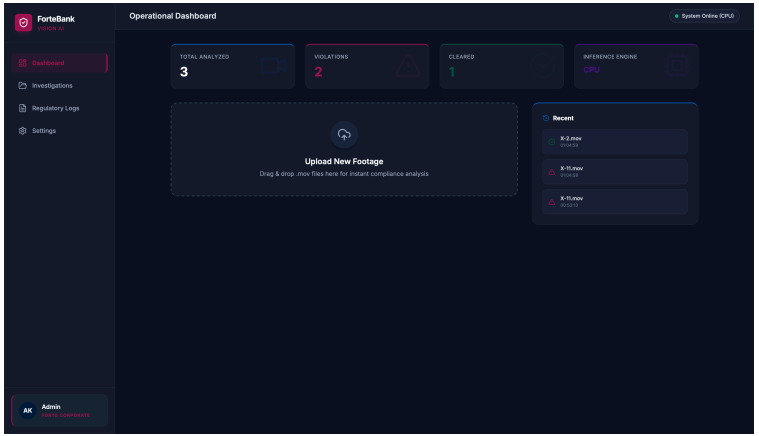
- 95% CI on F1 is 45–76%. The lower bound overlaps the trivial baseline.
- Required sample size for ±5% CI at 95% confidence is ~400 clips. We had 42.
- Treat every number on subsequent slides as an existence proof, not a performance claim.

# What we shipped: the verifiable subset

*One rule. Deterministic. Cited. Runs on a branch server, offline.*

## THE RULE

- Detects: cashier leaves workstation zone for more than 1.2s
- Cites: NBRK Resolution 231, Article 22, verified on adilet.zan.kz
- Output: timestamp, evidence JPEG, article number, direct URL

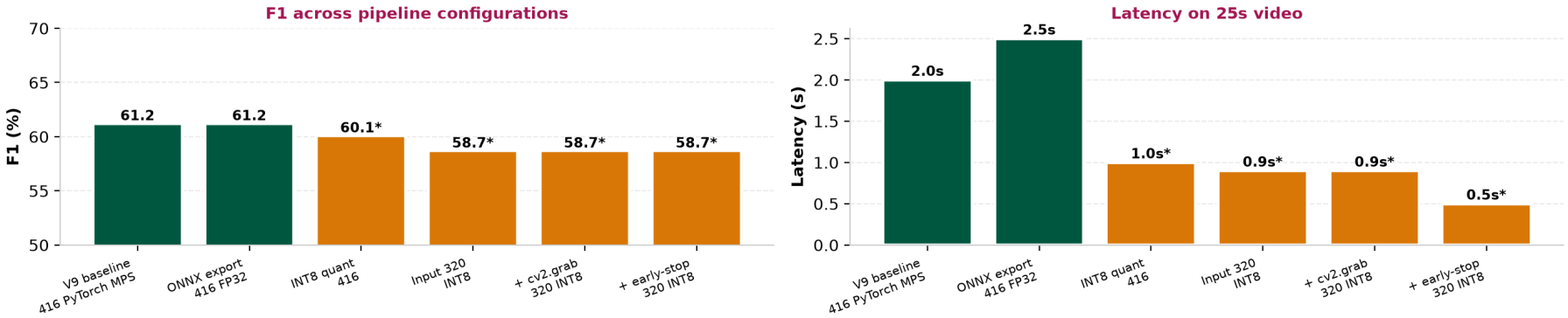



## THE SYSTEM

- 1.0s per 25s video on M1 CPU, 179 MB peak RAM
- Docker image, fully offline after pull
- Live demo, working API, deterministic output

# Ablation: what we measured vs. what we estimated

*The shipped configuration has NOT been re-benchmarked on all 42 clips.*



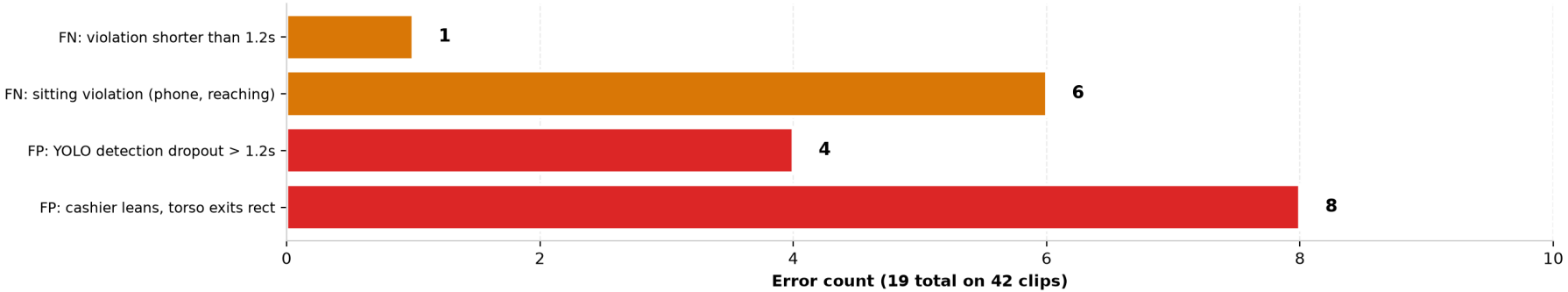

*Green = measured on 42-clip benchmark. Amber (*) = estimated, not re-run.*

## HONEST READING

- Full F1 benchmark was run on V9 PyTorch MPS only. Post-ONNX numbers are estimated.
- The shipped INT8 320x320 model was spot-checked on X-11 (correct) and X-48 (correct).
- Re-running the benchmark on the shipped model is the first task for Phase 2.

# Failure taxonomy: 19 errors, 4 root causes

*Not 'the model is bad'. Specific failure modes with specific fixes.*

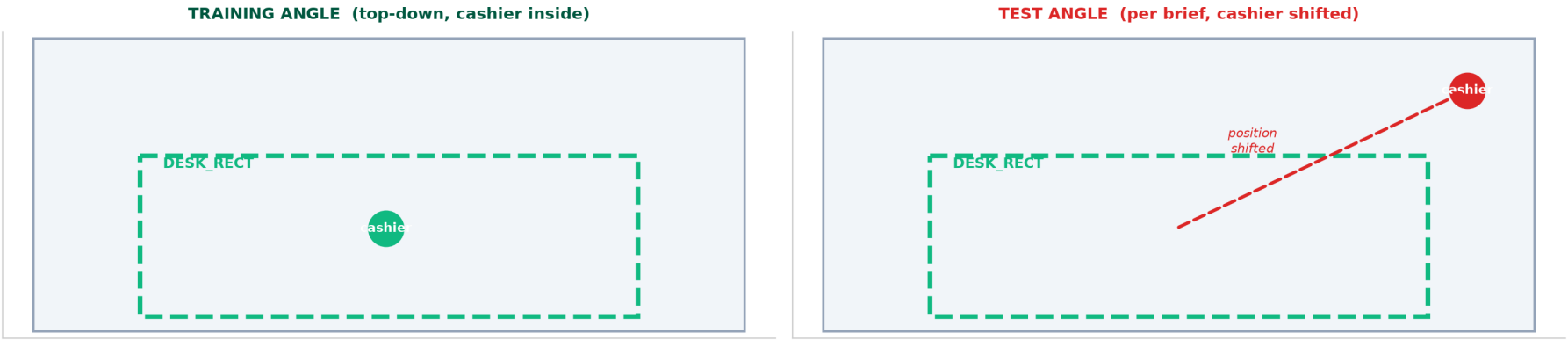



## WHAT IS FIXABLE VS. WHAT IS NOT

- 8 FPs from lean: tighten the vertical rect boundary. One line. May cost a few TPs.
- 4 FPs from detection dropout: require 2 consecutive frames of no-detection before firing.
- 6 FNs from sitting violations: needs an action classifier. Not fixable in 48h.
- 1 FN from short violation: lower streak threshold from 6 to 4. Risk: more FPs.

# Generalization: the biggest honest limitation

*The dataset brief says test is filmed from a different angle. We did not solve this.*




## TWO VIABLE PATHS, BOTH OUT OF 48H SCOPE

- Approach A: per-video desk detection using YOLO's 'dining table' class + homography
- Approach B: manual desk-corner annotation at inference time (UI prompt before analysis)
- Both require 1–2 days of implementation and validation. Deferred to Phase 2.

# What we rejected: with proper caveats

*Two single tests. We do not claim these generalize to the categories.*

### MOONDREAM2 (VLM)

- Prompt: 'Is there a mobile phone visible?' (single-sentence, no chain-of-thought)
- n=4 clean videos. All 4 produced false positives.
- Root cause: IP desk phone on every desk. VLM answered 'yes'.
- n=2 real violations. Both answered 'no'. Phone too small.

### YOLOV8S-POSE

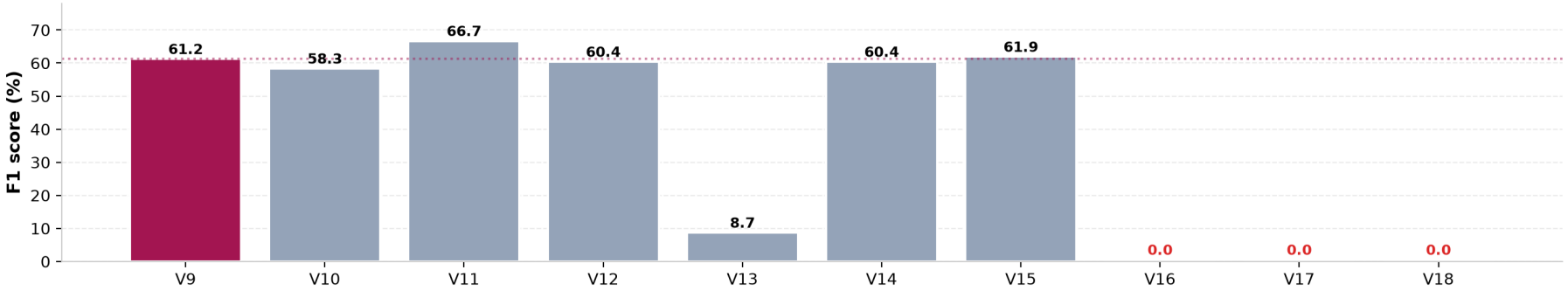
- Head/wrist keypoints as proxy for presence and phone use
- Failure 1: keypoints drop for 1-3 frames on lean-forward (12/12 FPs)
- Failure 2: chin-rest and phone-call are geometrically identical in 2D
- No threshold separates them at this resolution

### CAVEAT

- Each test was a single experiment. n=4 and n=12 respectively. Single prompt, single model.
- Not tried: alternative prompts, chain-of-thought, or other VLMs. That is the study a real researcher would run.
- We do not claim 'VLMs fail'. We claim 'this VLM with this prompt failed'.

# The iteration story: measurement as ground truth

*9 versions. 3 produced 0% F1. Each looked correct before measurement.*




## WHAT BROKE AND WHY

- V13 (central column): box expanded to 70% of frame → everything counted as 'at desk' → 8.7% F1
- V16 (adaptive home base): cashier never exceeded threshold → streak never fired → 0% F1
- V17 (no-null-detection guard): guard prevented streak from building → 0% F1
- V18 (y2=1.0): box covered bottom of frame, same failure as V17 → 0% F1

# Related work and the honest gap

*We do not beat SOTA. We characterize the achievable floor under constraint.*

| Approach | Resource requirement | Why not used here |
|---|---|---|
| AVA, MOMA, SlowFast | 100k+ labeled videos, 8-128 GPUs | Not applicable: 42 videos, 1 M1 |
| Retail fraud (Chen et al. 2019) | 10k+ labeled transactions | Not public, not obtainable in 48h |
| Pose-based action recognition | Requires stable keypoints | Fails at this resolution (slide 8) |
| VLM zero-shot (GPT-4V, Llama-Vision) | Cloud APIs or 7 GB local models | Prohibited by data residency |
| This work | 42 clips, 48h, CPU-only, offline | 61.22% F1, verifiable, deployable |

## POSITIONING

- This is not SOTA. It is 'what can you ship today under the constraints above'.
- The gap to SOTA is order-of-magnitude on data and compute. Not closeable in 48h.
- The contribution is a feasibility bound and a legal-anchoring pattern, not a new model.

# The ask: statistical power, not features

*What would move this from pilot study to validated claim.*

| | | |
|---|---|---|
| **WEEKS 1 TO 6** | Label 400 clips (from 42). Two annotators. | *Moves CI from ±15% to ±5%* |
| **WEEKS 7 TO 10** | Fine-tune 3D action classifier for sitting violations | *Target: +5 F1 on phone/reaching* |
| **WEEKS 11 TO 12** | Per-video desk detection for angle invariance | *Target: generalizes to test angle* |
| **WEEKS 13 TO 16** | One pilot branch, 30 days live footage | *Success: 80% catch, <10% FP* |

**WHAT WE NEED FROM YOU**

- 400 historical clips for labeling · one pilot branch contact · nothing else

# This is a pilot study.

We proved: deterministic, legally-anchored, CPU-deployable CV is achievable in 48h.
We did not prove: generalization, statistical significance, or action recognition.
**The gap is measurable, not hidden.**

**github.com/almansur04/forte-cash-cv**

```
docker run -p 8000:8000 almansur04/forte-cash-api:latest
```

API key: forte-hack-2026